

## Economics G6410 - Mathematical Methods for Economists - Fall 2017

Math Camp Exam

6:10 - 8:00 p.m., September 5, 2017

This is a closed-book, closed-note exam, and no calculator is allowed. There are 4 questions in total, and the full grade is 80 points. Good luck :)

1. (20 pts) Consider a  $2 \times 2$  matrix

$$A = \begin{bmatrix} 1 & -1 \\ 1 & -\frac{3}{2} \end{bmatrix}$$

(a) (5 pts) Find the eigenvalues of this matrix  $A$ .

$\det(\lambda I - A) = \det\left(\begin{bmatrix} \lambda - 1 & 1 \\ -1 & \lambda + \frac{3}{2} \end{bmatrix}\right) = (\lambda - 1)\left(\lambda + \frac{3}{2}\right) + 1 = (\lambda + 1)\left(\lambda - \frac{1}{2}\right)$ . The eigenvalues of  $A$  are  $\lambda_1 = -1$ ,  $\lambda_2 = \frac{1}{2}$ .

(b) (7 pts) Find a  $2 \times 2$  invertible matrix  $P$  and a  $2 \times 2$  diagonal matrix  $\Lambda$  s.t.  $P^{-1}AP = \Lambda$ .

Solving  $(\lambda_i I - A)\mathbf{x} = \mathbf{0}$ ,  $i = 1, 2$ , we get one eigenvector corresponding to  $\lambda_1 = -1$ :  $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ ;

and one eigenvector corresponding to  $\lambda_2 = \frac{1}{2}$ :  $\mathbf{x}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ .

Let  $P = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ , it's straightforward to verify that  $P$  is invertible.

Let  $\Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$ .

Then we have  $AP = A \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{bmatrix} = \begin{bmatrix} A\mathbf{x}_1 & A\mathbf{x}_2 \end{bmatrix} = \begin{bmatrix} \lambda_1 \mathbf{x}_1 & \lambda_2 \mathbf{x}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{bmatrix} \Lambda = P\Lambda \implies P^{-1}AP = \Lambda$ .

(c) (4 pts) What is  $A^n$  ( $A$  raised to the  $n$ -th power)? (Write out its 4 entries explicitly.)

$$P^{-1}AP = \Lambda \implies A = P\Lambda P^{-1}.$$

$$P^{-1} = \begin{bmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{bmatrix}.$$

$$\begin{aligned}
A^n &= (P\Lambda P^{-1})^n = P\Lambda^n P^{-1} \\
&= \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} (-1)^n & 0 \\ 0 & \left(\frac{1}{2}\right)^n \end{bmatrix} \begin{bmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{bmatrix} \\
&= \begin{bmatrix} (-1)^n & \left(\frac{1}{2}\right)^{n-1} \\ 2 \cdot (-1)^n & \left(\frac{1}{2}\right)^n \end{bmatrix} \begin{bmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{bmatrix} \\
&= \begin{bmatrix} \frac{(-1)^{n-1}}{3} + \frac{1}{3} \cdot \left(\frac{1}{2}\right)^{n-2} & \frac{2 \cdot (-1)^n}{3} - \frac{1}{3} \left(\frac{1}{2}\right)^{n-1} \\ \frac{2 \cdot (-1)^{n-1}}{3} + \frac{1}{3} \left(\frac{1}{2}\right)^{n-1} & \frac{4 \cdot (-1)^n}{3} - \frac{1}{3} \left(\frac{1}{2}\right)^n \end{bmatrix}.
\end{aligned}$$

(d) (**4 pts**) Let's view a  $2 \times 2$  real matrix as a 4-dimensional real vector and use the Euclidean distance  $d_2$  to measure the distance between two such matrices. Does the sequence  $(A^n)$  converge to some limit? If so, what is the limit?

$(A^n)$  converges in  $(\mathbb{R}^4, d_2)$  iff every entry of  $A^n$  converges in  $(\mathbb{R}, d_2)$ . It can easily be seen that none of its entry converges because of the fluctuating term  $(-1)^n$ .

Take  $(A^n)_{11} = \frac{(-1)^{n-1}}{3} + \frac{1}{3} \cdot \left(\frac{1}{2}\right)^{n-2}$  for example. We can see that for  $\forall n \geq 10$ ,  $d((A^n)_{11}, (A^{n+1})_{11}) \geq \frac{1}{10}$ , so  $((A^n)_{11})$  is not a Cauchy sequence and hence not convergent.

2. (**20 pts**) Please provide a TRUE or FALSE answer to each of the following statements, and give a proof or a counter-example (**4pts** each).

(a) Suppose  $(W, +, \cdot)$  and  $(U, +, \cdot)$  are both vector subspaces of vector space  $(V, +, \cdot)$ , then  $(W \cup U, +, \cdot)$  is also a vector subspace of  $(V, +, \cdot)$ .

FALSE. Take  $V = \mathbb{R}^2$ ,  $W := \left\{ \alpha \mathbf{e}_1 : \mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \alpha \in \mathbb{R} \right\}$  and  $U := \left\{ \beta \mathbf{e}_2 : \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \beta \in \mathbb{R} \right\}$ .

$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \in W \subset W \cup U$ ,  $\begin{bmatrix} 0 \\ 1 \end{bmatrix} \in U \subset W \cup U$ . However  $\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \notin W \cup U$ .

(b) Consider a metric space  $(X, d)$  where  $d$  is the discrete metric:  $\forall x, y \in X$

$$d(x, y) = \begin{cases} 1, & \text{if } x \neq y \\ 0, & \text{if } x = y \end{cases}$$

Then a set  $S \subset X$  is compact if and only if it is finite.

TRUE. " $\Leftarrow$ " is straightforward.

$\Rightarrow$ : For any infinite set  $S \subset X$ ,  $\{\{x\}\}_{x \in S}$  is an open cover of  $S$  under the discrete metric, for which we cannot find a finite subcover. Therefore  $S$  is not compact.

Notice that  $\forall x \in S$ ,  $\{x\} = B_{\frac{1}{2}}(x) \subset S$  is an open set under the discrete metric. (As we have discussed in class, in fact any set in  $X$  is both closed and open under the discrete metric.)

(c) Suppose  $f : X \rightarrow Y$  is discontinuous at  $x \in X$  and  $g : Y \rightarrow Z$  is a continuous function, then  $g \circ f : X \rightarrow Z$  is discontinuous at  $x$ .

FALSE. Take  $f : \mathbb{R} \rightarrow \mathbb{R}$  defined as  $f(x) = \begin{cases} 1, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases}$  ;  $g : \mathbb{R} \rightarrow \mathbb{R}$  defined as  $g(x) =$

$1, \forall x \in \mathbb{R}$ . Then  $g \circ f(x) = 1, \forall x \in \mathbb{R}$ .  $g \circ f$  is continuous.

(d) Assume  $\max_{\mathbf{x} \in \mathbb{R}^n} u(\mathbf{x})$  s.t.  $\mathbf{p} \cdot \mathbf{x} \leq w$  has a solution for any  $(\mathbf{p}, w)$  in some convex set  $S \subset \mathbb{R}^n \times \mathbb{R}$ . The value function  $v : S \rightarrow \mathbb{R}$  defined as  $v(\mathbf{p}, w) := \max_{\mathbf{x} \in \mathbb{R}^n} \{u(\mathbf{x}) \text{ s.t. } \mathbf{p} \cdot \mathbf{x} \leq w\}$  is a quasi-convex function.

TRUE. For any  $(\mathbf{p}, w), (\mathbf{p}', w') \in S, \lambda \in [0, 1]$ , WTS  $v(\lambda \mathbf{p} + (1 - \lambda) \mathbf{p}', \lambda w + (1 - \lambda) w') \leq \max\{v(\mathbf{p}, w), v(\mathbf{p}', w')\}$ .

Notice that  $(\lambda \mathbf{p} + (1 - \lambda) \mathbf{p}') \cdot \mathbf{x} \leq \lambda w + (1 - \lambda) w'$  implies either  $\mathbf{p} \cdot \mathbf{x} \leq w$  or  $\mathbf{p}' \cdot \mathbf{x} \leq w'$ , then any feasible point under  $(\lambda \mathbf{p} + (1 - \lambda) \mathbf{p}', \lambda w + (1 - \lambda) w')$  must be feasible under either  $(\mathbf{p}, w)$  or  $(\mathbf{p}', w')$  and therefore either  $v(\lambda \mathbf{p} + (1 - \lambda) \mathbf{p}', \lambda w + (1 - \lambda) w') \leq v(\mathbf{p}, w)$  or  $v(\lambda \mathbf{p} + (1 - \lambda) \mathbf{p}', \lambda w + (1 - \lambda) w') \leq v(\mathbf{p}', w')$ .

(e) Every compact-valued, closed-graph correspondence is upper hemicontinuous.

FALSE. Take  $F : \mathbb{R} \rightarrow \mathbb{R}$  defined as  $F(x) = \begin{cases} \left\{\frac{1}{x}\right\}, & \text{if } x \neq 0 \\ \{0\}, & \text{if } x = 0 \end{cases}$ . Then  $F$  is compact-valued,

closed-graph but not continuous and hence not uhc at  $x = 0$ . In fact, we need local boundedness, a stronger notion than compact-valuedness for cgp to imply uhc.

3. (20 pts) A consumer's utility maximization problem is

$$\begin{aligned} \max_{(x_1, x_2) \in \mathbb{R}_+^2} \quad & \ln x_1 + x_2 \\ \text{s.t.} \quad & x_1 + px_2 \leq m \quad (1) \\ & x_1 \geq 1 \quad (2) \end{aligned}$$

where  $p > 0, m \geq 1$  are parameters.

(a) (2 pts) Does there exist a solution to the maximization problem?

The constraint set  $D(p, m) := \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1 + px_2 \leq m, x_1 \geq 1\}$  is non-empty, closed and bounded in  $(\mathbb{R}^2, d_2)$ , and thus compact for any  $p > 0, m \geq 1$ ;  $f(x_1, x_2) = \ln x_1 + x_2$  is continuous in  $(x_1, x_2)$ . By Weierstrass Thm the problem has a solution for any  $p > 0, m \geq 1$ .

(b) (2 pts) Argue that the budget constraint (1) must hold with equality.

Suppose not, and  $m - x_1 - px_2 = \varepsilon > 0$  at an optimum. Then we could increase  $x_1$  by  $\varepsilon$ , satisfy all the constraints, and obtain a strictly higher value of the objective function because  $t \mapsto \ln t$  is strictly increasing, a contradiction.

(c) (5 pts) Write the Lagrangian. State the Kuhn-Tucker necessary conditions for a maximum. Are these conditions sufficient for a maximum?

Rewrite the problem as

$$\begin{aligned}
& \max_{(x_1, x_2) \in \mathbb{R}^2} && \ln x_1 + x_2 \\
& \text{s.t.} && m - x_1 - px_2 \geq 0 \\
& && x_1 - 1 \geq 0 \\
& && x_2 \geq 0
\end{aligned}$$

$$\mathcal{L}(x_1, x_2, \lambda_1, \lambda_2, \lambda_3, m, p) = \ln x_1 + x_2 + \lambda_1 (m - x_1 - px_2) + \lambda_2 (x_1 - 1) + \lambda_3 x_2$$

FOCs:

$$\frac{\partial \mathcal{L}}{\partial x_1} = \frac{1}{x_1} - \lambda_1 + \lambda_2 = 0 \quad (0.1)$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = 1 - p\lambda_1 + \lambda_3 = 0 \quad (0.2)$$

Complementary Slackness Conditions:

$$m - x_1 - px_2 = 0, \lambda_1 \geq 0 \quad (0.3)$$

$$\lambda_2 (x_1 - 1) = 0, \lambda_2 \geq 0, x_1 - 1 \geq 0 \quad (0.4)$$

$$\lambda_3 x_2 = 0, \lambda_3 \geq 0, x_2 \geq 0 \quad (0.5)$$

Equations (0.1) to (0.5) are the Kuhn-Tucker necessary conditions for a maximum.

Since the objective function is concave and the constraint functions are linear, the Lagrangian is concave, therefore these conditions are also sufficient for a maximum.

(d) **(2 pts)** Are there any feasible points where the constraint qualification fails? Are these points maximizers?

Note that  $\nabla_x (m - x_1 - px_2) = [-1, -p]$ ,  $\nabla_x (x_1 - 1) = [1, 0]$ ,  $\nabla_x (x_2) = [0, 1]$ .

If CQ fails it must be the case that all the 3 inequality constraints bind, i.e.  $x_1 = 1, x_2 = 0$ .

Then  $m - x_1 - px_2 = m - 1$ .

That is, only if  $m = 1$  we might consider  $(1, 0)$  as a feasible point where CQ fails. Actually when  $m = 1$ ,  $(1, 0)$  is the unique feasible point, so it is a maximizer.

(e) **(5 pts)** Solve for maximizer  $\mathbf{x}^* = (x_1^*, x_2^*)$  for any  $p > 0, m \geq 1$ .

Solve the K-T conditions. We need to discuss different ranges of  $(p, m)$  satisfying  $m > 1$ .

Notice that from (b) we know the first constraint must always bind, so we just need to discuss the rest two nonnegativity constraints.

**Case 1:** If both the 2nd and 3rd constraints bind, then we have  $x_1^* = 1, x_2^* = 0$ . Therefore  $m = 1$ , we are back to the non-CQ case we discuss in (d).

**Case 2:** Only the 3rd constraint binds:

$$\lambda_2^* = 0, x_2^* = 0. \quad x_1^* = m > 1, \lambda_1^* = \frac{1}{m} > 0, \lambda_3^* = \frac{p}{m} - 1.$$

K-T solution  $\exists!$  iff  $\lambda_3^* = \frac{p}{m} - 1 \geq 0 \Leftrightarrow m \leq p$ .

**Case 3:** Only the 2nd constraint binds:

$$\lambda_3^* = 0, x_1^* = 1. \quad x_2^* = \frac{m-1}{p} > 0, \lambda_1^* = \frac{1}{p} > 0, \lambda_2^* = \frac{1}{p} - 1.$$

K-T solution  $\exists!$  iff  $\lambda_2^* = \frac{1}{p} - 1 \geq 0 \Leftrightarrow 0 < p \leq 1$ .

**Case 4:** Both the 2nd and the 3rd constraints are slack:

$$\lambda_2^* = \lambda_3^* = 0. \quad \lambda_1^* = \frac{1}{p}, x_1^* = p, x_2^* = \frac{m}{p} - 1.$$

$$\text{K-T solution } \exists! \text{ iff } \begin{cases} x_1^* = p \geq 1 \\ x_2^* = \frac{m}{p} - 1 \geq 0 \end{cases} \Leftrightarrow 1 \leq p \leq m.$$

From (1) we know existence, so the K-T solutions above are the unique maximizers (at each  $(p, w)$ ).

(Do not forget the non-CQ point from (d):  $x_1^*(p, 1) = 1, x_2^*(p, 1) = 0$  when  $m = 1$ , which can be included in Case 2.)

$$\mathbf{x}^*(p, m) = \begin{cases} \left(p, \frac{m}{p} - 1\right) & \text{if } 1 < p \leq m \\ (m, 0) & \text{if } 1 \leq m < p \\ \left(1, \frac{m-1}{p}\right) & \text{if } 0 < p \leq 1 < m \end{cases}$$

(f) (**4 pts**) Find the value function  $v(p, m)$ . Verify the equation in the envelope theorem about the derivatives of  $v(p, m)$  w.r.t.  $p$  and  $m$  when  $m \geq p > 1$ .

$$v(p, m) = \ln x_1^* + x_2^* = \begin{cases} \ln p + \frac{m}{p} - 1, & \text{if } 1 < p \leq m \\ \ln m, & \text{if } 1 \leq m < p \\ \frac{m-1}{p}, & \text{if } 0 < p \leq 1 < m \end{cases}$$

When  $m \geq p > 1$ :

$$\begin{aligned} \frac{\partial v}{\partial p}(p, m) &= \frac{1}{p} - \frac{m}{p^2}; \quad \frac{\partial v}{\partial m}(p, m) = \frac{1}{p} \\ \frac{d\mathcal{L}}{dp}(x_1^*, x_2^*, \lambda_1^*, \lambda_2^*, \lambda_3^*, p, m) &= \frac{d}{dp}(\ln x_1^* + x_2^* + \lambda_1^*(m - x_1^* - px_2^*) + \lambda_2^*(x_1^* - 1) + \lambda_3^*x_2^*) = -\lambda_1^*x_2^* = \\ &= -\frac{1}{p} \cdot \left(\frac{m}{p} - 1\right); \\ \frac{d\mathcal{L}}{dm}(x_1^*, x_2^*, \lambda_1^*, \lambda_2^*, \lambda_3^*, p, m) &= \lambda_1^* = \frac{1}{p}. \end{aligned}$$

4. (**20 pts**) A one-good closed economy<sup>1</sup> has the following two equilibrium conditions:

$$\left(\frac{C_{t+1}}{C_t}\right)^{\frac{1}{\sigma}} = \beta(f'(K_{t+1}) + 1) \quad (3)$$

$$K_{t+1} + (1 + \tau)C_t = f(K_t) + K_t \quad (4)$$

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<sup>1</sup>Assume population growth rate and depreciation rate are both zero.

where  $\sigma \geq 1$  is the constant elasticity of intertemporal substitution,  $\beta \in (0, 1)$  is the discount factor,  $\tau > 0$  is the constant consumption tax rate.  $C$  denotes consumption and  $K$  capital. The production function  $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$  is  $C^2$ ,  $f' > 0$ ,  $f'' < 0$ .

(a) (**3 pts**) Characterize the steady state  $(K^*, C^*)$  of the model (that is, write down the equations that characterize  $K^*$  and  $C^*$  such that  $K_{t+1} = K_t = K^*$ ,  $C_{t+1} = C_t = C^*$ ).

Plug  $K_{t+1} = K_t = K^*$ ,  $C_{t+1} = C_t = C^*$  in equations (3) and (4) we have

$$\begin{aligned} 1 &= \beta (f'(K^*) + 1) \\ (1 + \tau) C^* &= f(K^*) \end{aligned}$$

Rearrange terms we have

$$\begin{aligned} \beta f'(K^*) + \beta - 1 &= 0 \\ (1 + \tau) C^* - f(K^*) &= 0 \end{aligned}$$

(b) (**8 pts**) Assume  $f(K) = K^\alpha$ ,  $\alpha \in (0, 1)$ . Log-linearize equations (3) and (4) around the steady state point  $(K_t, K_{t+1}, C_t, C_{t+1}) = (K^*, K^*, C^*, C^*)$ .

(That is, write both equations in the form  $\widehat{F}(K_t, K_{t+1}, C_t, C_{t+1}) = 0$ ; define the log-deviations  $\widehat{K}_t := \ln K_t - \ln K^*$ ,  $\widehat{K}_{t+1} := \ln K_{t+1} - \ln K^*$ ,  $\widehat{C}_{t+1} := \ln C_{t+1} - \ln C^*$ ,  $\widehat{C}_t := \ln C_t - \ln C^*$ ; take a first order Taylor approximation to  $F$  in the log-deviations around the point  $(\widehat{K}_t, \widehat{K}_{t+1}, \widehat{C}_t, \widehat{C}_{t+1}) = (0, 0, 0, 0)$  and set the expression equal to zero. You can use the shortcuts we discussed in class.)

Plug in  $f(K) = K^\alpha$ , equation (3) becomes  $\left(\frac{C_{t+1}}{C_t}\right)^{\frac{1}{\sigma}} = \beta (\alpha K_{t+1}^{\alpha-1} + 1)$ ; equation (4) becomes  $K_{t+1} + (1 + \tau) C_t = K_t^\alpha + K_t$

LHS of Equation (3):  $\left(\frac{C_{t+1}}{C_t}\right)^{\frac{1}{\sigma}} = \frac{1}{\sigma} (\widehat{C}_{t+1} - \widehat{C}_t)$

RHS of Equation (3):  $\beta (\alpha K_{t+1}^{\alpha-1} + 1) = \beta (\alpha K_{t+1}^{\alpha-1} + 1) = \frac{\alpha(\alpha-1)(K^*)^{\alpha-1} \widehat{K}_{t+1}}{\alpha(K^*)^{\alpha-1} + 1}$

So the log-linearization of equation (1) is given as

$$\frac{\alpha(\alpha-1)(K^*)^{\alpha-1}}{\alpha(K^*)^{\alpha-1} + 1} \widehat{K}_{t+1} - \frac{1}{\sigma} (\widehat{C}_{t+1} - \widehat{C}_t) = 0$$

LHS of Equation (4):  $K_{t+1} + (1 + \tau) C_t = \frac{K^* \widehat{K}_{t+1} + (1 + \tau) C^* \widehat{C}_t}{K^* + (1 + \tau) C^*}$

RHS of Equation (4):  $K_t^\alpha + K_t = \frac{\alpha(K^*)^{\alpha-1} \widehat{K}_t + K^* \widehat{K}_t}{(K^*)^\alpha + K^*}$

So the log-linearization of equation (2) is given as

$$\frac{\alpha(K^*)^{\alpha-1} \widehat{K}_t}{(K^*)^\alpha + K^*} - \frac{K^*}{K^* + (1 + \tau) C^*} \widehat{K}_{t+1} - \frac{(1 + \tau) C^*}{K^* + (1 + \tau) C^*} \widehat{C}_t = 0$$

(c) (**9 pts**) Now consider the steady state characterizing equations you write down in (a). Do these equations implicitly define  $K$  and  $C$  as  $C^1$  functions of  $(\beta, \tau)$  in an open ball around  $(K^*, C^*)$ ? If

so, find the derivatives of these implicit functions with respect to both  $\beta$  and  $\tau$ .

$$\text{Let } F(K, C, \beta, \tau) = \begin{bmatrix} \beta f'(K) + \beta - 1 \\ (1 + \tau)C - f(K) \end{bmatrix}$$

Clearly  $F(K^*, C^*, \beta, \tau) = \mathbf{0}$  and  $F$  is  $C^1$  at  $(K^*, C^*, \beta, \tau)$ .

$$f'_{(K,C)}(K, C, \beta, \tau) = \begin{bmatrix} \frac{\partial F_1}{\partial K} & \frac{\partial F_1}{\partial C} \\ \frac{\partial F_2}{\partial K} & \frac{\partial F_2}{\partial C} \end{bmatrix} = \begin{bmatrix} \beta f''(K) & 0 \\ -f'(K) & 1 + \tau \end{bmatrix}$$

$$f'_{(K,C)}(K^*, C^*, \beta, \tau) = \begin{bmatrix} \beta f''(K^*) & 0 \\ -f'(K^*) & 1 + \tau \end{bmatrix}$$

$\det(f'_{(K,C)}(K^*, C^*, \beta, \tau)) = \beta(1 + \tau)f''(K^*) < 0$ , so  $f'_{(K,C)}(K^*, C^*, \beta, \tau)$  is invertible. By the Implicit Function Theorem, there exists an implicitly defined and  $C^1$  function  $g$  of  $(\beta, \tau)$  in an open ball around  $(K^*, C^*)$ .

$$[f'_{(K,C)}(K, C, \beta, \tau)]^{-1} = \frac{1}{\beta(1 + \tau)f''(K)} \begin{bmatrix} 1 + \tau & 0 \\ f'(K) & \beta f''(K) \end{bmatrix}$$

$$f'_{(\beta,\tau)}(K, C, \beta, \tau) = \begin{bmatrix} \frac{\partial F_1}{\partial \beta} & \frac{\partial F_1}{\partial \tau} \\ \frac{\partial F_2}{\partial \beta} & \frac{\partial F_2}{\partial \tau} \end{bmatrix} = \begin{bmatrix} f'(K) + 1 & 0 \\ 0 & C \end{bmatrix}$$

$$\begin{aligned} g'(\beta, \tau) = \begin{bmatrix} \frac{\partial K}{\partial \beta} & \frac{\partial K}{\partial \tau} \\ \frac{\partial C}{\partial \beta} & \frac{\partial C}{\partial \tau} \end{bmatrix} &= -[f'_{(K,C)}(K, C, \beta, \tau)]^{-1} f'_{(\beta,\tau)}(K, C, \beta, \tau) \\ &= -\frac{1}{\beta(1 + \tau)f''(K)} \begin{bmatrix} 1 + \tau & 0 \\ f'(K) & \beta f''(K) \end{bmatrix} \begin{bmatrix} f'(K) + 1 & 0 \\ 0 & C \end{bmatrix} \\ &= \begin{bmatrix} -\frac{f'(K) + 1}{\beta f''(K)} & 0 \\ -\frac{(f'(K) + 1)f'(K)}{\beta(1 + \tau)f''(K)} & -\frac{C}{1 + \tau} \end{bmatrix} \end{aligned}$$

(d) (**5 pts, bonus**) Give economic interpretations of your results in (c), using the definitions of the parameters  $\beta$  and  $\tau$ .

Notice that  $\frac{\partial K}{\partial \beta} > 0$ ,  $\frac{\partial C}{\partial \beta} > 0$ : a more patient economy has higher consumption and capital stock levels in the steady state, holding other parameters fixed.

$\frac{\partial K}{\partial \tau} = 0$ ,  $\frac{\partial C}{\partial \tau} < 0$ : a higher rate of consumption tax leads to a lower consumption level, holding other parameters fixed. But it does not affect the capital stock level in the steady state, as it is unrelated to the the marginal return to capital (when the asset market clears).